CEGE 4160/5180 - Course Syllabus Applied Machine Learning for Civil, Environmental, and Geo- Engineers

Class: 9:45AM - 11:00AM MW, CivE 202

This course is designed for graduate students (CEGE 5180) and senior undergraduate students (CEGE 4160) who want to apply machine learning to civil, environmental, and geo-engineering problems. It covers models from classical methods to current deep learning, each with one or two CEGE applications. Students build every lab and homework with a coding agent of their choice and are graded on problem framing, validation, and engineering judgment. After a one-week review of prerequisites, regression, classification, and PCA are taught as one-week foundations so that deep learning starts in week 6, before the project proposal is due.

Contents

1 Teaching Team **2**
- 1.1 Instructors . . . 2
- 1.2 Teaching Assistants . . . 2
- 1.3 Office Hours . . . 2

2 About the course **3**
- 2.1 Course Focus . . . 3
  - 2.1.1 Ultimate Aim . . . 3
  - 2.1.2 Foundational Aim (for building basic knowledge and understanding) . . . 3
  - 2.1.3 Mediating Aim (for labs, homework, and the project) . . . 3
- 2.2 Course Prerequisites . . . 4
- 2.3 Text Books . . . 4
- 2.4 Software and Coding Agents . . . 5
- 2.5 Course Web Page . . . 5
- 2.6 Grading . . . 5
- 2.7 Tentative Outline . . . 6

3 Guidelines **7**
- 3.1 Assignments . . . 7
  - 3.1.1 Labs and Homework . . . 7
  - 3.1.2 Final Project . . . 8
  - 3.1.3 Submissions . . . 9
  - 3.1.4 Late Submission Policy . . . 9
- 3.2 Use of Generative AI . . . 9

1 Teaching Team

1.1 Instructors

Dr. Seongjin Choi

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Office Hours:	TBD

Dr. Choi received the B.S., M.S., and Ph.D. degree in the Department of Civil and Environmental Engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, in 2015, 2017, and 2021, respectively. Dr. Choi's research interests are broad and interdisciplinary, encompassing Urban Mobility Data Analytics, Spatiotemporal Data Modeling, Deep Learning & Artificial Intelligence, and Connected Automated Vehicles (CAV) & Cooperative-ITS. He is particularly driven by the desire to optimize urban mobility and contribute to the development of a sustainable and efficient urban transportation system. His works involve utilizing data analytics to draw valuable insights from urban mobility data and applying cutting-edge AI technologies in the field of transportation.

1.2 Teaching Assistants

TBD

email:	TBD
Office Hours:	TBD

1.3 Office Hours

- `https://choi-seongjin.github.io/contact/`

2 About the course

2.1 Course Focus

Civil, Environmental, and Geo-Engineering is rapidly becoming a data-rich field, with vast amounts of data generated from various sources such as structural sensors, traffic sensors, remote sensing, and massive simulation-based models. This course is designed for graduate students and senior undergraduate students who are eager to leverage machine learning techniques for applications in civil, environmental, and geo-engineering.

The course provides a comprehensive overview of machine learning models, spanning from classical methods to state-of-the-art deep learning approaches, tailored specifically for CEGE problems. Each week, specific topics will be addressed through a combination of theoretical lectures and practical examples. The course emphasizes hands-on learning, with one or two example applications in the civil, environmental, and geo-engineering context for each topic.

You will use a coding agent (Claude Code, GitHub Copilot, Codex, or another of your choice) in every lab and homework: the goal is to learn how to *apply* a method to an engineering problem and to *validate* what the agent produced, not to hand-implement every algorithm. In three weeks (4, 10, and 12) the instructor replays their own agent-built application in class before you extend it.

2.1.1 Ultimate Aim

By the end of this course, students will be able to understand the basic concepts of machine learning, build machine learning models with a coding agent, and check the results to solve real-world problems in civil, environmental, and geo-engineering.

2.1.2 Foundational Aim (for building basic knowledge and understanding)

1. Learn the basic concepts of machine learning and deep learning, including how a model is trained with gradient descent
2. Learn the main models (linear and logistic regression, PCA, neural networks, CNNs, RNNs, graph neural networks, autoencoders, and physics-informed machine learning) and when to use each
3. Learn common mistakes in machine learning, such as overfitting and data leakage
4. Learn the tools: Python, NumPy, pandas, PyTorch, and a coding agent

2.1.3 Mediating Aim (for labs, homework, and the project)

1. Build machine learning models on real CEGE datasets with a coding agent
2. Check every result and fix the mistakes the agent makes
3. Evaluate and compare models, and analyze the results
4. Complete an individual project, from proposal to final report

2.2 Course Prerequisites

The course uses the full first- and second-year mathematics sequence that every CSE engineering major completes. Civil, environmental, and geoengineering students finish it by the end of their sophomore year.

- **Multivariable calculus** (Calculus III): Math 2374 (CSE) or Math 2263, or equivalent. Partial derivatives, gradients, and the multivariable chain rule are how every model in this course is trained: gradient descent from Week 3 and backpropagation from Week 6.
- **Linear algebra and differential equations** (Calculus IV): Math 2373 (CSE) or Math 2243, or equivalent. Matrix products, rank, and eigenvalues are the language of every network, and Week 5 builds the singular value decomposition on top of eigenvalues; ordinary differential equations return in physics-informed machine learning (Week 14).
- **Probability and statistics**: CEGE 3102 (Uncertainty and Decision Analysis), or an equivalent statistics course. Random variables, expectation, the Gaussian distribution, and maximum likelihood.
- **Numerical methods and programming**: CEGE 3101 (Computer Applications I), or equivalent experience in Python (or another language you can move to Python quickly). You should be able to write a function, a loop, and a plot without help, and to solve a problem numerically when there is no closed form: iterate until convergence, check the error, and know when a numerical answer is wrong. Gradient descent is exactly this kind of method.

Calculus I and II (Math 1371/1372 or 1271/1272) are prerequisites of the courses above and are assumed. Graduate students from other programs meet these through equivalent coursework; if you are unsure, talk to the instructor in Week 1. HW00 (released in Week 2) checks these prerequisites directly, so you know where you stand before the deep learning weeks begin.

2.3 Text Books

There is **NO** official textbook required in this course.

While there is no official textbook, the instructor will share relevant reading materials as needed throughout the course. There are four optional textbooks for further reference, which are freely available online:

- *Probabilistic Machine Learning for Civil Engineers* by James-A. Goulet[1]
- *Probabilistic Machine Learning An Introduction* by Kevin Patrick Murphy[2]
- *Probabilistic Machine Learning: Advanced Topics* by Kevin Patrick Murphy[3]
- *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control* by Steven L. Brunton and J. Nathan Kutz[4]

[1] `https://profs.polymtl.ca/jagoulet/Site/Goulet_web_page_BOOK.html`
[2] `https://probml.github.io/pml-book/book1.html`
[3] `https://probml.github.io/pml-book/book2.html`
[4] `https://databookuw.com/`

2.4 Software and Coding Agents

All labs and homework run on your own laptop in VS Code with a terminal: Python (NumPy, pandas, matplotlib, scikit-learn, PyTorch) and the Jupyter extension, installed from the course's `requirements.txt` (posted on Canvas with the setup guide). Coding agents live in the editor and the terminal, so that is where this course lives; there is no hosted notebook service. Labs and homework are sized to run on a laptop CPU in minutes; no GPU is needed. Nothing needs to be purchased.

> Every student must have a coding agent of their own by the end of Week 1. Which one is up to you (Claude Code, GitHub Copilot, OpenAI Codex, or another agent that can edit notebooks and run code), free student tier or paid. The course does not provide licenses.

A setup guide with the current student offers is posted on Canvas. Lab00 (take-home, Week 1, pass/fail) is complete when Python and VS Code with the Jupyter extension run a notebook cell on your laptop, you have a GitHub account with one repository, and your chosen agent is installed and signed in from the VS Code terminal. Lab01 (Week 2, in class) is the agent's first build: it loads a CSV file by URL, produces a labelled plot, and commits the work to your repository; you submit the transcript with the AI-use disclosure cell.

2.5 Course Web Page

The course web site will be used for messages and announcements, data transfer and code sharing, assignments, forums, questionnaires, and to browse your currents scores and standing in the course. The course web site is available through the University of Minnesota *Canvas* server. If you are unfamiliar with *Canvas* I recommend that you run through the University of Minnesota's "Canvas orientation" at

- Canvas: Getting Started for Students

To access the course web site go to

- `https://canvas.umn.edu/`

login, and go to the CEGE 4160/5180 site. Make it a habit to look-over this site at least a once a day.

2.6 Grading

The course grade has three components. Weights are fixed and differ slightly between the two course numbers. There is no exam and no quiz.

Component	CEGE 4160	CEGE 5180	Notes
Labs and homework	45%	40%	HW00 and Lab02–Lab07, equal weight; Lab00 and Lab01 are pass/fail
Final project	50%	55%	proposal 5, checkpoint 5, presentation with live demo and oral check 15, report 25 (4160) or 30 (5180)
Participation	5%	5%	pitches, peer review, attendance

Part C. Every lab and homework has three parts. Parts A and B are the same for everyone. Part C (about 20% of the assignment: a from-scratch implementation, a derivation, or a small paper reproduction) is required for CEGE 5180 students and counts as extra credit for CEGE 4160 students; the labs and homework category is capped at 100%.
Rubrics. Each assignment is released with its rubric. Every criterion is scored 1–5 with written anchors at 5, 3, and 1, and the score is the weighted sum. Validation and engineering judgment (baseline comparison, sanity checks, one documented failure case, one agent mistake you caught) is never weighted below 25%.
Does it run. A notebook that does not run top to bottom, or an app that does not launch at the demo, caps every technical criterion at 2 of 5. Every submission ends with an AI-use disclosure cell (Section 3.2).

2.7 Tentative Outline

The following is the planned outline for the class (Spring 2027: classes Wed 1/20 to Mon 5/03; no class Mon 1/18; spring break 3/08–3/12; finals 5/06–5/12). Reality can deviate from it; Canvas is authoritative.

Week	Dates		Topic
1	–	1/20	Orientation
2	1/25	1/27	Review of prerequisites: statistics, linear algebra, optimization, Python/NumPy/pandas **- Lab01: coding-agent setup and first build** **- HW00: prerequisite problem set**
3	2/01	2/03	Regression as a one-layer network: gradient descent vs. closed form, maximum likelihood, regularization, validation and spatial splits
4	2/08	2/10	Classification as a one-layer network: logistic and softmax regression, cross-entropy, evaluation metrics **- Lab02: regression and classification on the same problem (urban tree canopy data: predict surface temperature, then classify heat hot spots)**

5	2/15	2/17	Unsupervised learning I: principal component analysis and the SVD (dimensionality reduction; autoencoders return to this in week 13) **- Lab03: SVD and PCA for sensor anomaly detection**
6	2/22	2/24	Deep learning I: multilayer perceptrons and backpropagation **- Project pitch (2 minutes, in class)**
7	3/01	3/03	Deep learning II: training neural networks **- Lab04: from logistic regression to an MLP on the Lab02 data, and debugging a broken training run** **- Project proposal**
8	3/08	3/10	*Spring Break*
9	3/15	3/17	Convolutional neural networks
10	3/22	3/24	Object detection and semantic segmentation **- Lab05: CNN on concrete crack images and object detection on traffic camera images**
11	3/29	3/31	Sequence models: RNN, LSTM, attention, Transformers **- Project checkpoint**
12	4/05	4/07	Spatiotemporal prediction and graph neural networks **- Lab06: sequence and graph-based traffic forecasting**
13	4/12	4/14	Unsupervised learning and generative models
14	4/19	4/21	Physics-informed machine learning **- Lab07: autoencoder on SHM sensors and PIML for fluid dynamics**
15	4/26	4/28	**Final project presentations and live demos**
16	5/03	–	AI Ethics; course wrap-up **- Final project report**

3 Guidelines

3.1 Assignments

3.1.1 Labs and Homework

There are six graded labs (Lab02–Lab07), in weeks 4, 5, 7, 10, 12, and 14. A lab starts in class on Wednesday and whatever you do not finish becomes that week's homework: you complete the same notebook at home and submit it by the Monday before the next lab (due dates are on Canvas). There is no separate homework set on top of a lab. Each lab starts with about 25 minutes on the internals of the week's method without an agent, followed by an applied part in which you use your agent on a CEGE dataset. In weeks 4, 10, and 12

the instructor first replays an application they built with a coding agent (the initial prompt, the mistakes the agent made, how they were caught), and the applied part is to extend it. Wednesdays without a lab are lecture, pitches, or the project checkpoint.

> The labs are designed to be HARD. They use real datasets drawn from current research (for example, Lab02 predicts urban surface temperature, and classifies heat hot spots, from tree-canopy data collected by Dr. Xue Feng's group), and you are expected to use a coding agent on them.

HW00, released in Week 2, is the one stand-alone homework: a problem set that checks the prerequisites (statistics, linear algebra, optimization, and programming). Lab00 (Week 1, take-home: installs and accounts) and Lab01 (Week 2, in class: agent setup) are pass/fail. HW00 and Lab02–Lab07 are graded on the same 1–5 rubric and weigh the same.

Each graded assignment has Part A (build and validate), Part B (analyze and report), and Part C (required for 5180, extra credit for 4160). The rubric is published with the assignment. You submit the completed notebook, run top to bottom before saving, with all figures generated by your code and an AI-use disclosure cell at the end (Section 3.2).

If any aspect of an assignment is unclear, ask during office hours or on the course forum; unclear instructions are a bug we want to fix.

3.1.2 Final Project

The final project is **individual**. You will design a machine learning project on a CEGE problem that interests you. Using a dataset you already have (for example, from your research) is encouraged; if you are unsure about topics or data, consult the TA or the instructor.

> Schedule at least one meeting with the TA or the instructor before the proposal is due, to check feasibility and scope.

Pitch (2 minutes, in class) — Week 6, Wed 2/24
One slide: the engineering question, the data, and what "working" would look like.

Proposal (max 2 pages) — Due Fri 3/05
Research question, dataset (with evidence that it loads), approach, and timeline. The proposal must name a linear or logistic regression baseline *and* a multilayer-perceptron baseline. A project that ends with a linear model is acceptable only if it beats the MLP baseline on validation data or argues why it wins. CEGE 5180 proposals add a related-work plan (at least five papers).

Checkpoint — Wed 3/31
The pipeline runs end to end and reports both baselines: the linear model and the MLP. CEGE 5180 students add an ablation plan.

Presentation and live demo (15 minutes) — Week 15, Mon 4/26 and Wed 4/28
A concise presentation of objectives, methods, results, and findings, followed by a live demo and an oral check. In the demo the instructor runs your app or notebook on an input of their choosing. In the oral check, with your agent closed, you explain one piece of your own

code and make a small change to it on the spot; this is the one moment in the course where you work without an agent, and it stands in for an exam. Presentations are peer-reviewed.

Final report (max 10 pages) — Due Mon 5/10
A written report structured like a conference submission (for example, a TRB Annual Meeting paper): methodology, data, results, validation, and conclusions. CEGE 5180 reports add a related-work section and an ablation or a technical novelty; a 4160 student who delivers these earns extra credit.

3.1.3 Submissions

ALL work in this class is submitted electronically through Canvas.

- Notebooks are submitted as ".ipynb" files, run top to bottom (Run All in the VS Code notebook toolbar) before saving, with outputs included.
- Written reports (proposal, final report) are submitted as ".pdf" files. *Google Docs*, *Microsoft Word*, and LaTeX can all export a ".pdf".
- If you need to submit multiple files, combine them into a single ".zip" file.

3.1.4 Late Submission Policy

- A **30-minute grace period** is granted for late submissions caused by system errors, computer malfunctions, or incorrect file attachments.
- Requests for a **48-hour extension** are permitted with a minor late submission penalty of typically 5% deducted from the total points. Such requests must be communicated to the instructor at least 24 hours before the deadline.
- Failure to submit an assignment by the deadline without prior arrangement for a 48-hour extension will result in a late submission penalty of typically 10% deducted from the total points.
- Assignments not submitted **within 48-hours** of the deadline will receive zero points.

3.2 Use of Generative AI

AI tools, including coding agents and chat assistants, may be used **without restriction** on homework, labs, and the project: for code, figures, analysis, and written text. Using them well is a skill this course teaches. Two rules apply everywhere, and one exception.

1. You are responsible for every result and every sentence you submit. A number you cannot reproduce, a fabricated citation, or a claim the data does not support is graded as your own error. Work that does not run caps every technical criterion at 2 of 5.
2. Disclose. Every submission ends with an AI-use disclosure cell: which tools you used, what they did (for example, "wrote the training loop", "drafted Section 3"), what you verified yourself, and a short prompt-log excerpt (your first prompt and one mistake the agent made that you caught).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and Grammarly for language assistance. After utilizing these tools, the author(s) thoroughly reviewed and revised the content as necessary and take full responsibility for the final version of the published article.

Figure 1: Example of an AI-use statement

For example:

There is no exam. The one place you work without your agent is the oral check at the final demo (Section 3.1.2), where you explain and change a piece of your own code on the spot. It draws on the same material as the labs and homework, so you must be able to explain every line you submit during the semester.